

Driving impactful AI initiatives at the national level

📁 EMPLOYMENT

U.S. National Science Foundation

Senior Advisor for Artificial Intelligence 2025–present
Directorate for Computer and Information Science and Engineering (CISE)
Program Director (Federal), CISE 2023–present
Program Director (IPA), CISE 2019–2022

George Mason University

Associate Professor, Department of Computer Science 2020–2023

Catholic University of America

Associate Professor, Department of Electrical Engineering and Computer Science 2016–2020
Assistant Professor, Department of Electrical Engineering and Computer Science 2010–2016

Johns Hopkins University

Adjunct Assistant Professor, Department of Computer Science 2010–2012

🎓 EDUCATION

Johns Hopkins University *Postdoctoral Researcher in Computer Science* 2008–2010

Rice University *Ph.D. in Computer Science* 2002–2008

Clarkson University *M.S. in Computer Science* 2000–2002

SUNY Fredonia *B.S. in Computer Science* 1996–2000

👥 LEADERSHIP AND PROGRAM MANAGEMENT

NSF CISE Senior Advisor for Artificial Intelligence 2025–present

Serves as the senior expert and strategic advisor on AI within the CISE Directorate, guiding research directions, policy development, and cross-directorate and cross-agency AI initiatives

NSF AI Steering Committee (Co-Chair) 2025–present

Co-chairs the NSF AI Steering Committee, providing strategic leadership and coordination across NSF directorates to shape national AI research priorities and initiatives

NSF National AI Research Institutes Program (Co-Lead) 2023–present

*Co-leads NSF's flagship AI program, managing a **\$640M** investment in 29 AI Institutes to accelerate high-impact research and applications across sectors via collaboration with academia, government, and industry*

NITRD AI R&D Interagency Working Group (Co-Chair) 2025–present

Leads interagency coordination of federal AI R&D efforts across 32 agencies, driving strategic initiatives to strengthen U.S. leadership in AI

NSF Robust Intelligence (Program Director)	2019–2022, 2023–present
<i>Manages core research program driving advancements in AI, machine learning, computer vision, human language technologies, and computational neuroscience</i>	
NITRD Intelligent Robotics and Autonomous Systems IWG (Co-Chair)	2021–2022
<i>Led interagency coordination of R&D across 28 federal agencies, accelerating innovation in resilient and efficient robotics and autonomous systems</i>	
Foundational Research in Robotics (Co-Lead)	2020–2022
National Robotics Initiative (Co-Lead)	2019–2022
<i>Led NSF's flagship robotics initiatives, supporting groundbreaking research to advance next-generation robotics and autonomous systems</i>	
International Partnerships	
<i>Advanced international partnerships in AI and robotics between NSF and Japan's Science and Technology Agency and UK's Engineering and Physical Sciences Research Council</i>	
Strategic Leadership & Team Alignment	
<i>Proven ability to inspire, lead, and align cross-functional teams toward a unified vision, driving successful outcomes through clear goal-setting, collaborative execution, and fostering a culture of innovation</i>	
Executive Communication & Stakeholder Engagement	
<i>Proven ability to convey complex technical concepts to a broad spectrum of stakeholders, from technical experts to executive leadership, ensuring clear understanding and alignment for informed decision-making</i>	

SELECTED AWARDS

NSF Special Acts Award	2025
NSF Director's Award for Superior Accomplishments	2021
Finalist – Best Application Paper	2021
<i>McMahon J and Plaku E: "Dynamic Multi-Goal Motion Planning with Range Constraints for Autonomous Underwater Vehicles Following Surface Vehicles." <i>IEEE International Conference on Automation Science and Engineering</i> pp. 704–711</i>	
Best Robotics Paper	2017
<i>Le D and Plaku E: "Cooperative Multi-Robot Sampling-Based Motion Planning with Dynamics." <i>International Conference on Planning and Scheduling</i> pp. 513–521</i>	
Best Student Paper	2015
<i>Wells A and Plaku E: "Adaptive Sampling-Based Motion Planning for Mobile Robots with Differential Constraints." <i>Springer LNCS: Towards Autonomous Robotic Systems</i> vol. 9287, pp. 283–295</i>	
Kaman Excellence in Research Award	<i>School of Engineering, Catholic University of America</i> 2015
Faculty Research Fellowship	<i>Office of Naval Research, U.S. Naval Research Laboratory</i> 2014
Burns Fellowship	<i>School of Engineering, Catholic University of America</i> 2011



RESEARCH AND TECHNICAL ACCOMPLISHMENTS

Robotics & Autonomous Systems

Motion Planning & Control: Led innovations in AI-integrated motion planning deployed across robotic domains to enable fast, efficient, and adaptive autonomy

Task-Oriented Autonomy: Advanced task and motion planning methods that empower robots to autonomously execute high-level missions with minimal supervision

Multi-Robot Coordination: Developed scalable coordination strategies for autonomous teams, enabling collaborative decision-making in dynamic, real-world settings

Natural Language Integration: Leveraged large language models to enable robots to understand and act on human intent expressed in natural language

AI & Machine Learning

AI-Driven Decision-Making: Designed scalable decision systems combining search, planning, and reasoning to optimize complex, high-impact operations

Generative AI & LLMs: Integrated large language models to enhance knowledge access, automate reasoning, and strengthen human-AI collaboration

Deep Learning & Reinforcement Learning: Applied deep and reinforcement learning to enable real-time adaptation and performance optimization in dynamic environments

Uncertainty & Probabilistic Modeling: Advanced AI robustness through probabilistic reasoning and Bayesian inference for reliable decision-making under uncertainty

AI Scalability: Scaled complex AI solutions using distributed computing frameworks to support large-scale planning, search, and learning

Industry Applications & Cross-Domain Impact

Defense: Delivered AI-powered autonomy across air, land, sea, and undersea platforms, enhancing mission success, resilience, and operational efficiency

Healthcare & Medical Systems: Applied AI in robotic-assisted surgery to improve precision, training efficacy, and patient safety in high-stakes clinical environments

Manufacturing & Logistics: Enabled intelligent mobility and coordination in warehouse automation, driving cost reduction, scalability, and supply chain optimization

Publications

100+ papers in AI and robotics conferences and journals, full list: erionplaku.github.io

Teaching

Taught undergraduate and graduate courses in AI, ML, Robotics, Data Mining, and Algorithms during academic tenure



INTERESTS

Chess

1982–present

Former professional chess player with national and international competition experience; currently a recreational player with a Chess.com Blitz rating of 2475

Soccer

1981–present

Longtime soccer player with active participation in organized recreational leagues